

Regularization

$$L(w) = \underbrace{\frac{1}{N} \sum_{i=1}^N L_i(b(x_i, w), y_i)}_{\text{Data Loss}} + \underbrace{\lambda R(w)}_{\text{Regularization}}$$

Data Loss

Regularization

- prevents the model from doing too well on training data.

$$L2: R(w) = \sum_k \sum_l w_{k,l}^2$$

$$L1: R(w) = \sum_k \sum_l |w_{k,l}|$$

$$L_i = -\log\left(\frac{e^{s_{y_i}}}{\sum_j e^{s_j}}\right) \quad \text{Softmax}$$

Optimization

$$\frac{df(x)}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$L = \frac{1}{N} \sum_{i=1}^N L_i + \sum_k W_k^2$$

$$L_i = -\log\left(\frac{e^{s_{y_i}}}{\sum_j e^{s_j}}\right)$$

$$s = b(x; w) = wx$$

want $\nabla_w L$

For one sample

$$L = -\log\left(\frac{e^{s_y}}{\sum_j e^{s_j}}\right)$$

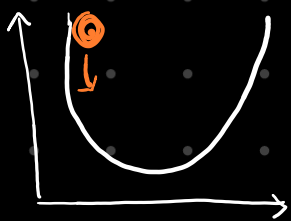
$$s = wx$$

while True:

weight_grad = evaluate_gradients(loss_fn, data, weights)

weights += - step-size * weight_grad.

Momentum Optimization



$v \rightarrow$ velocity

$$w_{t+1} = w_t - v_t$$

$$v_t = \beta * v_{t-1} + \nabla b(x_t)$$

- memory of past gradients.



$$V_{t+1} = pV_t + \nabla F(x_t)$$

$$x_{t+1} = x_t - \alpha V_{t+1}$$

- $V_0 = 0$

- l.r. = α

- momentum = p

- gradients at steps $g_0, g_1, g_2, \dots$

①

$$V_1 = pV_0 + g_0 = g_0$$

$$x_1 = x_0 - \alpha V_1 = x_0 - \alpha g_0$$

②

$$V_2 = pV_1 + g_1 = pg_0 + g_1$$

$$x_2 = x_1 - \alpha V_2 = (x_0 - \alpha g_0) - \alpha (pg_0 + g_1)$$

$$= x_0 - \alpha g_0 - \alpha pg_0 + \alpha g_1$$

$$= x_0 - \alpha (1+p)g_0 - \alpha g_1$$

③

$$V_3 = pV_2 + g_2$$

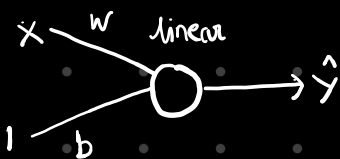
$$V_3 = p(pg_0 + g_1) + g_2 = p^2g_0 + pg_1 + g_2$$

$$x_3 = x_2 - \alpha V_3$$

Adagrad $\rightarrow$ Adaptive learning rate.

"sparse"

- vector is sparse when most of entries are zero.



$$W_{n+1} = W_n - \eta \frac{dL}{dW}$$

$$\frac{dL}{dW} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dW} = 2(x - \hat{y}) \cdot x$$

Formulation:

(idea is that we vary the l.r. across parameters)

$$V_t = V_{t-1} + (\nabla W_t)^2 \quad \} \text{ past gradients sum (squared)}$$

$$W_{t+1} = W_t - \frac{\eta \nabla W_t}{\sqrt{V_t + \epsilon}}$$



Disadvantage

- $\eta \rightarrow$ decreases
- as you progress through epochs, V_t term starts becoming very large $\Rightarrow$ the adaptive l.r " $\underline{\eta}$ " becomes very small.
$$\frac{\eta}{\sqrt{V_t} + \epsilon}$$

RMS Prop $\rightarrow$ Root Mean Squared Propagation

$$V_t = \beta V_{t-1} + (1-\beta)(\nabla W_t)^2 ; \beta = 0.95$$

$$W_{t+1} = W_t - \frac{\eta}{\sqrt{V_t} + \epsilon} (\nabla W_t)$$

$$V_0 = 0$$

$$V_1 = 0.95 \times 0 + 0.05 (\nabla W_1)^2$$

$$V_2 = 0.95 \times 0.05 (\nabla W_1)^2 + 0.05 \times (\nabla W_2)^2$$

$$V_3 = 0.95 \times 0.95 \times 0.05 (\nabla W_1)^2 + 0.95 \times 0.05 (\nabla W_2)^2 + 0.05 \times (\nabla W_3)^2$$

Adam optimizer

$$W_{t+1} = W_t - \frac{\eta}{\sqrt{V_t} + \epsilon} \times m_t \quad \text{where}$$

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) \nabla W_t \longrightarrow \text{momentum}$$

$$V_t = \beta_2 V_{t-1} + (1-\beta_2) (\nabla W_t)^2 \longrightarrow \text{RMS Prop}$$

$$\hat{m}_t = \frac{m_t}{1-\beta_1^t}, \quad \hat{V}_t = \frac{V_t}{1-\beta_2^t}$$

